XML Descriptions for Integration Template

> Yeastar P-Series Cloud Edition allows you to create CRM/Helpdesk integration templates, where you can define the necessary settings, including specific API endpoints, data fields, and functional scenarios required for the integration. This topic provides detailed description of the setting elements in the XML templates.

Template conventions

The template follows the **XML 1.1 standard** specification. This section describes the syntax and conventions of the template.

Variable

The template supports the use of variables for dynamic data. When the template is executed, the variables referenced within the template will be replaced with their respective values.

The following table shows the source of variable values and how to reference the variables in the template.

Item	Description

Source	Variables in this template can be obtained form the following sources:
	• Pre-defined variables in the PBX system
	• Parameter values input by users during the integration
	• Values retrieve from the request response
	You can retrieve values from the response body by specifying a JSON path, as shown in the examples below.
	Tip: For more information about the path syntax, see GJSON Path Syntax.
	◦ To retrieve a response value, use the following syntax:
	`<Output Name="UserUniqueId" Path="data.#.id"></Output>`
	◦ For retrieve a value that meets specific conditions, use following syntax:
	`<Output Name="BusinessNumber2" Path="identities.#(type=="phone_number")#|1.value"></Output>`

Item	Description
Reference format	Variables in this template follow the `text/template` syntax specification of Go Language, as shown below: • Standard variable: `{{.varname}}` • Conditional variable: `{{ if .varname }} Welcome, member! {{ else }} Please sign up. {{ end }}` • Function calls with variable: `{{ Capitalize .varname }}`

Function

The template supports the following function, which can be used in conjunction with variables to dynamically manipulating or formatting the variable value.

Function	Description
TimeFormat	The function can be used to customize the format of a timestamp. **Format**: {{ TimeFormat `Timestamp_variable` " `format_string` " " `Whether_to_convert_to_UTC_time` " }} **Example**: `{{ TimeFormat .StartTimeUnix "yyyy-MM-ddTHH:mm:ss.000Zz" "1" }}` Assuming the value of `{{.StartTimeUnix}}` is `1672531199`, the final output would be `2023-01-01T00:59:59.000Z`.
ToMillis	The function can be used to convert a seconds timestamp to a milliseconds timestamp. **Format**: {{ ToMillis `Timestamp_variable` }} **Example**: `{{ ToMillis .StartTimeUnix }}` Assuming the value of `{{.StartTimeUnix}}` is `1672531199`, the final output would be `1672531199000`.
Capitalize	The function can be used to capitalizes the first letter of a string. **Format**: {{ Capitalize `variable_string` }} **Example**: `{{ Capitalize .variable_string }}` Assuming the value of `{{.variable_string}}` is `hello world`, the final output would be `Hello world`.
UrlEncode	The function can be used to URL encode a string. **Format**: {{ UrlEncode `.variable` }} **Example**: `{{ UrlEncode .variable}}` Assuming the value of `{{.variable_string}}` is `hello world!`, the final output would be `hello%20world%21`.

Elements

The templates offers the following main elements that help define the integration interactions, ensuring that the template can seamlessly connect with the third-party system and process data based on integration requirements.

The following example shows the main structure of the template.

```
<Information>
  <Scenarios>
    <Scenario Id="1" Type="REST">
      <Parameters>
        <Parameter Name="example" Value="example"></Parameter>
        ...
      </Parameters>
      <Requests>
        <Request Name="ExampleRequest" Method="GET" URL="https://api.example.com/getuser">
           <Outputs>
            <Output Name="ContactId" Path="data.#.id" Type="string"></Output>
            ...
          </Outputs>
        </Request>
      </Requests>
    </Scenario>
  </Scenarios>
</Information>
```

Main Element	Description
Information	The `<Information>` element is a root element of the XML template, providing the basic information and configuration overview through its attributes. It also contains the `<Scenarios>` child element to describe the functional scenarios and integration capabilities of the template. For the details of the attributes, see Template property.
Scenarios	The `<Scenarios>` element provides a collection of `<Scenario>` child elements, which contains further child elements, such as `<Parameters>`, `<Requests>`, and `<Outputs>`, to describe specific functional scenarios for the integration. The PBX will load the configured scenarios and execute the HTTP requests or operations defined within it. The template supports the following scenarios. • Authentication scenario • User association scenario • Automatic contact synchronization scenario • Automatic contact creation scenario • Automatic ticket creation scenario (for Helpdesk system only) • Call journal scenario • Chat journal scenario

Main Element	Description
	Note: If the specified scenarios do not meet your requirements, you can manually edit the template configurations for supplementary usage. For more information, see Extended configurations.
Parameters	The `<Parameters>` element is used to define a collection of parameters that are associated with the scenarios or requests. These parameters typically include dynamic values or configuration settings required for the execution of the scenario. They can be passed in the request URL, body, or headers, depending on the use case.
Requests	This `<Requests>` element provides a collection of requests for the scenarios to implement specific features.
Outputs	The `<Outputs>` element contains a list of `<Output>` elements that define the data to be extracted from the API response.

Template property

The `<Information>` element provides the properties of the template, including the basic template information and configuration overview, as listed below:

Attribute	Description
Provider	The type of the template, with a fixed value of `crm` or `helpdesk`.
Name	The name of the template.
Key	Unique identifier for the template.
Logo	The filename of the logo.
Remark	A description or remark about the template.
Version	The version of the template. E.g. `1.0.0`.
AuthType	The authentication method used in the integration. • `none`: None authentication. • `basic`: Basic authentication. • `oauth2`: OAuth2 authentication. • `bearertoken`: Bearer Token authentication.

Attribute	Description
MaxConcurrentRequest	The maximum number of concurrent HTTP requests allowed to the CRM/Helpdesk.
UserAssociation	Whether the User Association feature is enabled or not. • `0`: Disabled. • `1`: Enabled.
CallJournal	Whether the Call Journal feature is enabled or not. • `0`: Disabled. • `1`: Enabled.
CreateNewContact	Whether the Create New Contact feature is enabled or not. • `0`: Disabled. • `1`: Enabled.
CreateNewTicket	Whether the Create New Ticket Automatically feature is enabled or not. • `0`: Disabled. • `1`: Enabled.
ChatJournal	Whether the Chat Journal feature is enabled or not. • `0`: Disabled. • `1`: Enabled.

Authentication scenario

This scenario defines the authentication method used by the CRM/Helpdesk system. The structure of the scenario and its key elements are outlined as below:

```
 <Scenario Id="AuthMethod" Type="AUTH">
   <Presets></Presets>
   <Parameters>
    <Parameter Name="AuthMethod" Value="oauth2"></Parameter>
    <Parameter Name="AuthEndPoint" Value="https://api.example.com/oauth/v2/auth?client_id={{.client_id}}">
</Parameter>
    <Parameter Name="TokenEndPoint" Value="https://api.example.com/oauth/v2/token?client_id=
{{.client_id}}&amp;client_secret={{.client_secret}}"></Parameter>
    <Parameter Name="ContentType" Value="application/json"></Parameter>
    <Parameter Name="AdditionalQueryString"></Parameter>
    <Parameter Name="Scope"
Value="contacts.read,contacts.write,calls.read,calls.write,user.read,user.write"></Parameter>
    <Parameter Name="CredentialType"></Parameter>
```

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```
    <Parameter Name="Base64EncodedCredential"></Parameter>
    <Parameter Name="TokenRefreshInterval" Value="25"></Parameter>
    <Parameter Name="TokenRefreshBody" Value="grant_type=refresh_token&amp;refresh_token=
{{.refresh_token}}"></Parameter>
    <Parameter Name="TokenRefreshEndpointMethod" Value="post"></Parameter>
    <Parameter Name="TokenRefreshEndpoint" Value="https://api.example.com/oauth2/token"></Parameter>
    <Parameter Name="TokenRefreshContentType" Value="application/x-www-form-urlencoded"></Parameter>
    <Parameter Father="CustomFieldList" Name="client_id" Editor="password" Title="Client ID" Value=""
IsDelete="0" IsRequired="1"></Parameter>
    <Parameter Father="CustomFieldList" Name="client_secret" Editor="password" Title="Client Secret"
Value="" IsDelete="0" IsRequired="1"></Parameter>
    <Parameter Father="CustomFieldList" Name="application_id" Editor="string" Title="Application ID"
Value="" IsDelete="0" IsRequired="1"></Parameter>
    <Parameter Father="CustomFieldList" Name="server_address" Editor="string" Title="Server Address"
Value="" IsDelete="0" IsRequired="1"></Parameter>
    <Parameter Father="PopulateTemplateString" Name="Content-Type" Type="Header" Value="application/json"
Description="The format of the request body"/>
    <Parameter Father="PopulateTemplateString" Name="Accept" Type="Header" Value="application/json"
Description="The expected response data format from the server "/>
    <Parameter Father="PopulateTemplateString" Name="X-CSRF-Token" Type="Header"
Value="f3e1a2b4c5d6e7f8g9h0" Description="Prevent Cross-Site Request Forgery (CSRF) attacks"/>
    <Parameter Father="PopulateTemplateString" Name="X-Client-Id" Type="Header" Value="{{.client_id}}"
Description="for client identification"/>
    <Parameter Father="FollowUpList" Name="ffb425cb-565d-49cd-92be-3290d0ab5129" Type="FollowUpRequest"
Value="" IsDelete="0">
        <Parameter Name="TriggeredEvent" Value="token_retrieved"></Parameter>
        <Parameter Name="RequestMethod" Value="post"></Parameter>
        <Parameter Name="Url" Value="https://api.example.com/v1/install/verify"></Parameter>
        <Parameter Name="Headers" Value="{"Content-Type":"application/json","X-Request-ID": "550e8400-e29b-
41d4-a716-446655440000","User-Agent":"YourApp-Integration/1.0.0"}"></Parameter>
        <Parameter Name="Body" Value="{"app_id":"your_unique_app_identifier","webhook_url":"https://your-
app.com/api/webhooks"}"></Parameter>
        <Parameter Name="ResponseFieldPath" Value=".data.access_token, .tokens[0].value"></Parameter>
        <Parameter Name="TargetVariable" Value="token,refresh"></Parameter>
    </Parameter>
    <Parameter Name="AuthMode" Value="pkce"></Parameter>
    <Parameter Name="CodeChallengeMethod" Value="S256"></Parameter>
    <Parameter Name="EnableKeepalive" Value="1"></Parameter>
    <Parameter Name="KeepaliveInterval" Value="5"></Parameter>
  </Parameters>
  <Requests></Requests>
 </Scenario>
```

Here is a detailed explanation of the specific elements within the scenario.

Scenario element

The `<Scenario>` element has the following attributes:

Attribute	Description
Id	The unique identifier of the scenario, the value should be fixed as `AuthMethod`.
Type	The type of the scenario, the value is `AUTH`, which means the scenario will execute authentication operations.

Parameter element

Attribute	Description
Name	The name of the authentication setting.
Value	The value of the authentication setting.
Father	The name of parent parameters. The authentication scenario includes three parent parameter types, each serving as a container to organize related child parameters into distinct setting groups. • `CustomFieldList`: Defines a set of fields for users to input authentication credentials (e.g., Client ID, Client Secret) in the configuration UI. **Note:** The variables captured in the custom fields are accessible for all the subsequent requests in the template. • `PopulateTemplateString`: Defines custom HTTP headers to be included in every predefined CRM / Helpdesk synchronization requests (e.g., user association, contact synchronization. etc.) sent to the CRM/Helpdesk system. • `FollowUpList`: Defines one or more automated API requests that are triggered after a specific event, such as after a token is successfully retrieved.

Parent parameter

Attribute	Description
CustomFieldList (for Authentication credentials input)	
Name	The variable name of the custom field, which can be reference in the template. For example, if you set the name to `client_id`, it can then be referenced as variable `{{.client_id}}`.
Editor	The type of the custom field. **Valid value**: • `string`: A standard input field where users can enter plain text. • `password`: A specialized input field for sensitive information, with text masked for privacy. • `list`: A selectable list that allows users to choose from a pre-defined list of options.
Title	The title of the custom field, which will be displayed above the custom field in the configuration UI.

Attribute	Description
Value	The pre-defined list of options. **Note:** • This attribute is available when the `Editor` is set to `list`. • Use commas to separate multiple options. For example, `option1,option2`.
IsRequired	Specify whether the field is mandatory. **Valid value**: • `0`: No. The field is optional and can be left blank. • `1`: Yes. The field is required. **Note:** This attribute is available when the `AuthMethod` is set to `oauth2`.
PopulateTemplateString (for Custom HTTP headers)	
Type	The type of the parameter, which should be fixed as `Header`.
Name	The name of the custom header.
Value	Header value, can be either a specific value or a variable. **Note:** The variable name in the authentication field can be used. For example, if a variable named `client_id` is set, you can reference it in the header value as `{{.client_id}}`.
Description	A brief explanation of the header's function or purpose for reference.
FollowUpList (for follow-up requests)	
Type	The type of the parameter, which should be fixed as `FollowUpRequest`.
Name	• For the parent parameter, the system will automatically generate an ID for it. • For the child parameters, this attribute can be used to fill in the name of the follow-up request settings.
Value	The value of the follow-up request settings.

Authentication settings

Setting	Description
AuthMethod	Authentication method. **Valid value**:

Setting	Description
	• `none`: This method is used when there is no authentication required, or when the CRM/Helpdesk only requires relevant variables for data exchange, such as an API Key or a webhook URL. • `basic`: This method authenticate requests using credentials, such as username and password or an API key. These credentials are be base64-encoded into a basic string and included in a header field of every HTTP request sent to the CRM/Helpdesk. • `oauth2`: This method uses tokens to grant access without sharing credentials. It requires obtaining the necessary header and other parameters (e.g. OAuth access token) using specific API requests to the CRM/Helpdesk. • `bearertoken`: This method authenticates requests using a bearer token. Every HTTP request sent to the CRM/Helpdesk must include a header field in the format of `Authorization: Bearer {{.AccessToken}}`, where `{{.AccessToken}}` is the token value granted by the CRM/Helpdesk.
AuthMode	Authorization mode. **Note:** This parameter is only available for `oauth2` authentication method. **Valid value**: • `authorization_code`: Standard authorization code flow. Use this mode when the CRM/Helpdesk expects a `client_secret` for authentication. • `pkce`: Authorization Code with PKCE (Proof Key for Code Exchange) flow. Use this mode when the CRM/Helpdesk supports PKCE authorization, as this provides an extra layer of security in addition to `client_secret`. In this mode, the client (i.e. PBX) generates a `code_verifier`, hashes it with SHA256 and encodes it with Base64URL to create a `code_challenge`, and sends this value to the authorization server upfront. The client stores the original `code_verifier` locally and presents it later during token exchange. The authorization server hashes the submitted `code_verifier` using the same method and compares it to the stored `code_challenge`. If they match, it proves that the client holds the original `code_verifier`, and only such a client can exchange the authorization code for an access token, effectively preventing interception attacks.
AuthEndPoint	The CRM/Helpdesk authorization URL that needs to be opened to initiate the authorization process. **Note:** This parameter is only available for `oauth2` authentication method.
TokenEndPoint	The CRM/Helpdesk API request URL for getting access token and refresh token. **Note:** This parameter is only available for `oauth2` and `bearertoken` authentication method.

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Setting	Description
ContentType	The content type of the token retrieval request. **Valid value**: • `application/json` • `application/x-www-form-urlencoded`
CodeChallengeMethod	The method used to transform the `code_verifier` into the `code_challenge`. **Note:** This parameter is only available when the `AuthMode` is set to `pkce`. **Valid value**: • `S256`: Hash the `code_verifier` with SHA256 and encode it with Base64URL to generate the `code_challenge` (recommended). • `plain`: Use the `code_verifier` directly as the `code_challenge` without hashing or encoding. Compatible with systems that do not support SHA256.
AdditionalQueryString	If the CRM/Helpdesk system authenticates requests using query parameters instead of scope, you can add the query variables in this parameter. **Note:** This parameter is only available for `oauth2` authentication method.
Scope	Specify the scope of CRM/Helpdesk data that the PBX is allowed to access. **Note:** This parameter is only available for `oauth2` authentication method.
CredentialType	The type of the credential. **Note:** This parameter is only available for `basic` authentication method. **Valid value**: • `username_password` • `api_key`
Base64EncodedCredential	Define the combination format of the credential variables, which will be used to generate the basic authentication string. **Note:** This parameter is only available for `basic` authentication method. E.g. `{{.username}}:{{.password}}` or `{{.api_key}}`.

Setting	Description
TokenRefreshEndpointMethod	The HTTP request method for the token refresh API. **Valid value**: • `get` • `post` **Note:** This parameter is only available for `oauth2` and `bearertoken` authentication method.
TokenRefreshEndpoint	The API request URL for refreshing access token. **Note:** This parameter is only available for `oauth2` and `bearertoken` authentication method.
TokenRefreshContentType	The content type of the token refresh request. **Valid value**: • `application/json` • `application/x-www-form-urlencoded` • `multipart/form-data` **Note:** This parameter is only available for `oauth2` and `bearertoken` authentication method.
TokenRefreshInterval	Define the time interval at which the token should be refreshed automatically (**Unit**: Minute). **Note:** This parameter is only available for `oauth2` and `bearertoken` authentication method.
TokenRefreshBody	The request body for the token refresh API. **Note:** This parameter is only available for `oauth2` and `bearertoken` authentication method.
EnableKeepalive	Whether to enable connection keep-alive. **Valid value**: • `0`: Disable. The system will not send keep-alive requests. • `1`: Enable. The system will send keep-alive requests to the CRM/Helpdesk at the specified interval.
KeepaliveInterval	Specify the interval for sending keep-alive request (in minutes).

Follow-up request settings

Setting	Description
TriggeredEvent	The event that triggers this follow-up request. **Valid value**: • `token_retrieved`: Initiate the follow-up request upon successful retrieval of an access token. • `token_refreshed`: Initiates the follow-up request upon successful refresh of an access token.
RequestMethod	The HTTP method for the API request. **Valid value**: • `get` • `post`
Url	The endpoint URL for the subsequent operation. For example, `https://api.example.com/rest-services/login`
Headers	The HTTP headers of the request (in JSON format). You can use `{{.varname}}` to insert dynamic values. For example, `{"Content-Type":"application/json", "X-Custom-Header": "{{.DynamicValue}}"}`
Body	The data payload of the request (in JSON format). You can use `{{.varname}}` to insert dynamic values. For example, `{"lastName":"{{.LastName}}","ownerId":"{{.Id}}","phone":"{{.BusinessNumber}}"}`
ResponseFieldPath	The JSON path to extract data from the response of the follow-up request. **Note:** Use commas `,` to separate multiple variable names. For example, `.data.access_token, .tokens[0].value`
TargetVariable	Define the name of the system variable to store the value extracted from the response, which can be existing variables or new custom variables. The variable value can be stored as global variables and referenced throughout the integration.

Setting	Description
	Note: • Use commas `,` to separate multiple variable names. For example, `token,refresh`. • The variables are assigned in the same order as the fields specified in the `ResponseFieldPath`. For example, the first variable receives the value from the first JSONPath, the second variable receives the value from the second JSONPath, and so on.

User association scenario

This scenarios is used to retrieve the list of users from the CRM/Helpdesk system. The structure of the scenario and its key elements are outlined as below:

```
<Scenario Id="UserAssociation" Type="REST">
   <Parameters></Parameters>
   <Requests>
    <Request Name="UserAssociation" Method="GET" ResponseType="application/json" RequestEncoding=""
URLFormat="https://www.api.example.com/v1/users?type=ActiveUsers">
    <Parameters>
            <Parameter Name="PaginationEnabled" Value="0" />
            <Parameter Name="PaginationType" Value="" />
            <Parameter Name="VarPageName" Value="page" />
            <Parameter Name="VarPageValue" Value="1" />
            <Parameter Name="VarPageSizeName" Value="page_size" />
            <Parameter Name="VarPageSizeValue" Value="10" />
            <Parameter Name="VarOffsetName" Value="from" />
            <Parameter Name="VarOffsetValue" Value="1" />
            <Parameter Name="VarLimitName" Value="limit" />
            <Parameter Name="VarLimitValue" Value="10" />
            <Parameter Name="VarNextPagePath" Value="links.next" />
     </Parameters>
     <Outputs>
      <Output Name="UserUniqueId" Path="users.#.id" Type=""></Output>
      <Output Name="FirstName" Path="users.#.First_Name" Type=""></Output>
      <Output Name="LastName" Path="users.#.Last_Name" Type=""></Output>
      <Output Name="Email" Path="users.#.Email" Type=""></Output>
     </Outputs>
    </Request>
   </Requests>
</Scenario>
```

Here is a detailed explanation of the specific elements within the scenario.

Scenario element

The `<Scenario>` element has the following attributes:

Attribute	Description
Id	The unique identifier of the scenario, the value should be fixed as `UserAssociation`.
Type	The type of the scenario, the value is `REST`, which means the scenario will execute a REST API using an HTTP request.

Request element

The `<Request>` element has the following attributes:

Attribute	Description
Name	The name of the request, which helps to identify the request within the scenario. The value should be fixed as `UserAssociation`.
Method	The HTTP method to be used for the request. In this scenario, the default HTTP method is `GET`, you can modify it if necessary.
ResponseType	The expected response format for the request, which should be in **JSON** format (`application/json`).
RequestEncoding	How the request should be encoded. In this scenario, the value is empty.
URLFormat	The request URL of the API endpoint.

Parameter element

The `<Parameter>` element provides has the following attributes:

Attribute	Description
Name	The name of the pagination settings used in the scenario.
Value	The value of the pagination settings.

Pagination settings

Setting	Description
PaginationEnabled	Whether to enable pagination. **Valid value**: • `0`: Disable. • `1`: Enable.

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Setting	Description
PaginationType	Specify the pagination mode. **Valid value**: • `page_based`: Retrieve user data based on page number and page size. • `offset_based`: Retrieve user data based on a starting index and record limit. • `link_based`: Retrieve user data by using the URL pointing to the next page of data in the API response.
VarPageName	Enter the parameter name used to specify the page number in API requests. **Note:** This parameter is only available for `page_based` pagination mode.
VarPageValue	Specify the initial page number from which to start querying data. **Note:** This parameter is only available for `page_based` pagination mode.
VarPageSizeName	Enter the parameter name used to specify the number of records per page in API requests. **Note:** This parameter is only available for `page_based` pagination mode.
VarPageSizeValue	Specify the maximum number of records to return in a single page. **Note:** This parameter is only available for `page_based` pagination mode.
VarOffsetName	Enter the parameter name used to specify the start position in API requests. **Note:** This parameter is only available for `offset_based` pagination mode.
VarOffsetValue	Specify the starting position for data retrieval by indicating the number of records to skip. For example, a value of `1` starts from the first record, while `100` starts retrieval from the 100th record. **Note:** This parameter is only available for `offset_based` pagination mode.
VarLimitName	Enter the parameter name used to specify the number of records per page in API requests. **Note:** This parameter is only available for `offset_based` pagination mode.

Setting	Description
VarLimitValue	Specify the maximum number of records to return in a single page. **Note:** This parameter is only available for `offset_based` pagination mode.
VarNextPagePath	Specify the JSON path (e.g. `links.next`) used to obtain the link pointing to the next page of data. **Note:** This parameter is only available for `link_based` mode.

Output element

The `<Output>` element has the following attributes:

Attribute	Description
Name	The name of the output variable, which represents the variables need to be retrieved from the response data.
Path	The location of the desired data field within the API response, using a path syntax to access nested data (e.g. `users.#.id`).
Type	The data type of the output variable. In this template, the type is fixed as `string`.

Variables

Variable	Description
UserUniqueId	The unique ID of the CRM/Helpdesk user.
FirstName	The first name of the user.
LastName	The last name of the user.
Email	The email of the user.

Automatic contact synchronization scenario

This scenarios is used to search for contacts in the CRM/Helpdesk system, and the returned information can be used to implement CRM contacts searching, automatic contact synchronization and call popup. The structure of the scenario and its key elements are outlined as below:

```
  <Scenario Id="SyncContactAuto" Type="REST">
   <Presets></Presets>
   <Parameters>
    <Parameter Name="ContactUrlType" Value="retrieve_from_contact"></Parameter>
    <Parameter Name="URLFormat"></Parameter>
    <Parameter Name="ContactFieldForUri" Value="data.#.contactUrl"></Parameter>
    <Parameter Name="ContactsIdEnable" Value="1"></Parameter>
    <Parameter Name="FirstNameEnable" Value="1"></Parameter>
    <Parameter Name="BusinessNumberEnable" Value="1"></Parameter>
    <Parameter Name="CustomValueEnable" Value="0"></Parameter>
   </Parameters>
   <Requests>
    <Request Name="Contacts" Method="GET" ResponseType="application/json" RequestEncoding=""
URLFormat="https://www.api.example.com/v1/Contacts/search?criteria=((Phone:equals:
{{.Phone}})or(Home_Phone:equals:{{.Phone}})or(Mobile:equals:{{.Phone}})or(Asst_Phone:equals:{{.Phone}}))">
     <Parameters></Parameters>
     <Outputs>
      <Output Name="ContactsId" Path="data.0.id" Type=""></Output>
      <Output Name="FirstName" Path="data.0.First_Name" Type=""></Output>
      <Output Name="BusinessNumber" Path="data.0.Phone" Type=""></Output>
      ...
      <Output Name="CustomValue" Path="data.0.Owner" Type=""></Output>
     </Outputs>
    </Request>
   </Requests>
  </Scenario>
```

Here is a detailed explanation of the specific elements within the scenario.

Scenario element

The `<Scenario>` element has the following attributes:

Attribute	Description
Id	The unique identifier of the scenario, the value should be fixed as `SyncContactAuto`.
Type	The type of the scenario, the value is `REST`, which means the scenario will execute a REST API using an HTTP request.

Parameter element

The `<Parameter>` element provides has the following attributes:

Attribute	Description
Name	The name of the configuration settings used in the scenario.
Value	The value of the configuration settings.

Configuration settings

Setting	Description
ContactUrlType	Specify the method for retrieving the contact URL for call popup. **Valid value**: • `specify_url_format`: Configure the call popup URL by specifying the URL format, where you can use variables to dynamically construct the URL. • `retrieve_from_contact`: Extract the contact URL directly from a specific field in the contact search response, where the contact URL can be retrieved.
URLFormat	The expression of the Contact URL. The following variables can be used to construct the URL. • The variables captured from the custom field inputs in the authentication process. • `ContactSyncType`: The contact type, which is retrieve from the name of the request used to search specific types of contact. • `ContactId`: The contact ID, which is retrieved from the response data in this scenario. • `CustomValue`: A custom field, which is retrieved from the response data in this scenario. For example, `{{.crm_url}}/crm/{{.CustomValue}}/tab/{{Capitalize .ContactSyncType}}/{{.ContactId}}`. **Note:** If the provided variables do not meet your requirements, you can define additional requests and retrieve the corresponding values for your desired variables. For more information, see Extended Configurations - Retrieve Additional Variables.
ContactFieldForUri	The JSON path used to retrieve the corresponding field for the contact URL from the response data. For example, `.data.#.contactUrl`.
ContactsIdEnable	Whether to retrieve contact's ID from the response data. The value should be set to `1`.
FirstNameEnable	Whether to retrieve the contact's first name from the response data. The value should be set to `1`.
LastNameEnable	Whether to retrieve the contact's last name from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.

Setting	Description
CompanyEnable	Whether to retrieve the contact's company from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
EmailEnable	Whether to retrieve the contact's email from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
BusinessNumberEnable	Whether to retrieve the contact's business phone number from the response data. **Note:** At least one phone number is required. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
BusinessNumber2Enable	Whether to retrieve the contact's second business phone number from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
BusinessFaxNumberEnable	Whether to retrieve the contact's business fax number from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
MobileNumberEnable	Whether to retrieve the contact's mobile number from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
MobileNumber2Enable	Whether to retrieve the contact's second mobile number from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.

Setting	Description
HomeNumberEnable	Whether to retrieve the contact's home number from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
HomeNumber2Enable	Whether to retrieve the contact's second home number from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
HomeFaxNumberEnable	Whether to retrieve the contact's home fax number from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
OtherNumberEnable	Whether to retrieve the contact's other type of number (other than business number or home number) from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
ZipCodeEnable	Whether to retrieve the zip code from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
StreetEnable	Whether to retrieve the street from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
CityEnable	Whether to retrieve the city from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.

Setting	Description
StateEnable	Whether to retrieve the state/province from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
CountryEnable	Whether to retrieve the country from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
RemarkEnable	Whether to retrieve the remark for the contact from the response data. **Valid value**: • `0`: Do not retrieve the data. • `1`: Retrieve the data.
CustomValueEnable	Whether to enable a custom variable, which can be used to construct the Contact's URL. **Valid value**: • `0`: Disable. • `1`: Enable.

Request element

The `<Request>` element has the following attributes:

Attribute	Description
Name	The name of the request, which helps to identify the request within the scenario. **Note:** The value should be the actual name of the specific contact type to be searched in the CRM/Helpdesk system, and will be referenced as a `ContactSyncType` variable.
Method	The HTTP method to be used for the request. In this scenario, the default HTTP method is `GET`, you can modify it if necessary.
ResponseType	The expected response format for the request, which should be in **JSON** format (`application/json`).
RequestEncoding	How the request should be encoded. In this scenario, the value is empty.
URLFormat	The request URL of the API endpoint.

Output element

The `<Output>` element has the following attributes:

Attribute	Description
Name	The name of the output variable, which represents the variables need to be retrieved from the response data.
Path	The location of the desired data field within the API response, using a path syntax to access nested data (e.g. `users.#.id`).
Type	The data type of the output variable. In this template, the type is fixed as `string`.

Variables

> **Note:** The following variable values retrieved from a call can be referenced across modules through out the call until it ends.

Variable	Description
ContactsId	**(Required)** The unique ID of the contact.
FirstName	**(Required)** The first name of the contact.
LastName	The last name of the contact.
Company	The company of the contact.
Email	The email of the contact.
BusinessNumber	The business phone number of the contact. **Note:** At least one phone number is required.
BusinessNumber2	The second business phone number of the contact.
BusinessFaxNumber	The business fax number of the contact.
MobileNumber	The mobile number of the contact.
MobileNumber2	The second mobile number of the contact.
HomeNumber	The home number of the contact.
HomeNumber2	The second home number of the contact.

Variable	Description
HomeFaxNumber	The home fax number of the contact.
OtherNumber	The other type of phone number of the contact.
ZipCode	Zip code.
Street	Street.
City	City.
State	State or province.
Country	Country.
Remark	Remark.
CustomValue	A custom variable, which is used to construct the Contact's URL.

Automatic contact creation scenario

This scenarios is used to create a new contact in the CRM/Helpdesk system when the caller's number does not match any existing contact. The structure of the scenario and its key elements are outlined as below:

```
 <Scenario Id="CreateNewContact" Type="REST">

   <Parameters></Parameters>
   <Requests>
    <Request Name="Contacts" Method="POST" ResponseType="application/json" RequestEncoding=""
URLFormat="https://www.api.example.com/v1/Contacts">
     <Parameters>
      <Parameter Name="Data" Type="Body" Value="{"data": [{"Last_Name":"{{.LastName}}","First_Name":"
{{.FirstName}}","Phone":"{{.BusinessNumber}}";}]}"></Parameter>
     </Parameters>
     <Outputs></Outputs>
    </Request>
   </Requests>
  </Scenario>
```

Here is a detailed explanation of the specific elements within the scenario.

Scenario element

The `<Scenario>` element has the following attributes:

Attribute	Description
Id	The unique identifier of the scenario, the value should be fixed as `CreateNewContact`.
Type	The type of the scenario, the value is `REST`, which means the scenario will execute a REST API using an HTTP request.

Request element

The `<Request>` element has the following attributes:

Attribute	Description
Name	The name of the request, which helps to identify the request within the scenario. **Note:** The value should be the actual name of the specific contact type to be created in the CRM/Helpdesk system.
Method	The HTTP method to be used for the request. In this scenario, the HTTP method is `POST`.
ResponseType	The expected response format for the request, which should be in **JSON** format (`application/json`).
RequestEncoding	How the request should be encoded. In this scenario, the value is empty.
URLFormat	The request URL of the API endpoint.

Parameter element

The `<Parameter>` element nested within the `<Request>` element defines the parameters included in the request body.

Attribute	Description
Name	The name of the parameter, which should be fixed as `Data`.
Type	The type of the parameter, which should be fixed as `Body`.
Value	Defines the required parameters to be passed through the request body, within a `data` structure. The parameter values can be retrieved from the pre-defined variables in the PBX system. **Note:** The format should follows XML parameter standards. For example, if a line break is required, you must use the escape character `
` to represent a newline.

Variables

Note: Only required variables are listed here. If you need additional parameters and variables, see Extended Configurations - Retrieve Additional Variables.

Variable	Description
Variables obtained from the `AuthMethod` scenario	The variables captured from the authentication credentials input during the authentication process of integration.
Target variables obtained from the follow-up requests	The target variables extracted from the responses of the follow-up requests.
{{.LastName}}	The last name of the contact.
{{.FirstName}}	The first name of the contact.
{{.BusinessNumber}}	The phone number of the contact.

Automatic ticket creation scenario (for Helpdesk system only)

This scenarios is used to create a new ticket in the Helpdesk system. The structure of the scenario and its key elements are outlined as below:

```
<Scenario Id="CreateNewTicket" Type="REST" >
   <Presets></Presets>
   <Parameters>
    <Parameter Name="EnableSubject" Value="1"></Parameter>
    <Parameter Name="Subject" Value="{{.Communication_Type}} {{.Call_Status}} - from {{.Call_From}} to
{{.Call_To}}"></Parameter>
    <Parameter Name="EnableDescription" Value="1"></Parameter>
    <Parameter Name="Description" Value="{{.Time}} {{.Communication_Type}} {{.Call_Status}} - from
{{.Call_From}} to {{.Call_To}} {{.Talk_Duration}}"></Parameter>
   </Parameters>
   <Requests>
    <Request Name="CreateNewTicket" Method="POST" Weight="0" ResponseType="application/json"
RequestEncoding="" URLFormat="https://{{.domain}}/api/v1/tickets">
     <Parameters>
      <Parameter Name="Data" Type="Body" Value="{"subject":"{{.Subject}}","contactId":"
{{.ContactId}}","phone":"{{.ContactNumber}}","description":"{{.Description}}";}"></Parameter>
     </Parameters>
     <Outputs></Outputs>
    </Request>
```

```
    </Requests>
</Scenario>
```

Here is a detailed explanation of the specific elements within the scenario.

Scenario element

The `<Scenario>` element has the following attributes:

Attribute	Description
Id	The unique identifier of the scenario, the value should be fixed as `CreateNewTicket`.
Type	The type of the scenario, the value is `REST`, which means the scenario will execute a REST API using an HTTP request.

Parameter element

Attribute	Description
Name	The name of the configuration settings used in the scenario.
Value	The value of the configuration settings.

Configuration settings

Settings	Description
EnableSubject	Whether to allow users to customize the subject of the ticket. **Valid value**: • `0`: Disable. • `1`: Enable.
Subject	Specify the default value of the subject. **Note:** The default value is `{{.Communication_Type}} {{.Call_Status}} - from {{.Call_From}} to {{.Call_To}}`. For more supported variables, see the parameter variable list.
EnableDescription	Whether to allow users to customize the description of the ticket. **Valid value**: • `0`: Disable. • `1`: Enable.

Settings	Description
Description	Specify the default value of the description. **Note:** The default value is `{{.Time}} {{.Communication_Type}} {{.Call_Status}} - from {{.Call_From}} to {{.Call_To}} {{.Talk_Duration}}`. For more supported variables, see the parameter variable list.

Request element

The `<Request>` element has the following attributes:

Attribute	Description
Name	The name of the request, which helps to identify the request within the scenario. The value should be fixed as `CreateNewTicket`.
Method	The HTTP method to be used for the request. In this scenario, the HTTP method is `POST`.
ResponseType	The expected response format for the request, which should be in **JSON** format (`application/json`).
RequestEncoding	How the request should be encoded. In this scenario, the value is empty.
URLFormat	The request URL of the API endpoint.

Parameter element

The `<Parameter>` element nested within the `<Request>` element defines the parameters included in the request body.

Attribute	Description
Name	The name of the parameter, which should be fixed as `Data`.
Type	The type of the parameter, which should be fixed as `Body`.
Value	Defines the required parameters to be passed through the request body. The parameter values can be retrieved from the pre-defined variables in the PBX system. **Note:** The format should follows XML parameter standards. For example, if a line break is required, you must use the escape character `
` to represent a newline.

Variables

XML Descriptions for Integration Template

Yeastar P-Series Cloud Edition - CRM / Helpdesk Integration Template

Note: Only required variables are listed here. If you need additional parameters and variables, see Extended Configurations - Retrieve Additional Variables.

Variable	Description
Variables obtained from the `AuthMethod` scenario	The variables captured from the authentication credentials input during the authentication process of integration.
Target variables obtained from the follow-up requests	The target variables extracted from the responses of the follow-up requests.
{{.Subject}}	The subject of the ticket. This variable value is retrieved from the `Subject` parameter.
{{.Description}}	The description of the ticket. The variable value is retrieved from the `Description` parameter.
{{.LinkedId}}	The unique ID of the call.
{{.Time}}	The time the call was made or received (in PBX system's time format).
{{.StartTimeUnix}}	The Unix timestamp of the call start time.
{{.EndTimeUnix}}	The Unix timestamp of the call end time.
{{.Call_From}}	The name and number of the calling party that initiated the call. **Format**: `Caller's name <Caller's number>`.
{{.CallerName}}	Caller's name.
{{.CallerNumber}}	Caller's number.
{{.CalleeName}}	Callee's name.
{{.CalleeNumber}}	Callee's number.
{{.CrmId}}	The unique ID of the CRM/Helpdesk user associated with a PBX extension.
{{.ExtensionFirstName}}	The first name of the extension user.
{{.ExtensionLastName}}	The last name of the extension user.

XML Descriptions for Integration Template

Yeastar P-Series Cloud Edition - CRM / Helpdesk Integration Template

Variable	Description
{{.ExtensionEmail}}	The email of the extension user.
{{.ContactSyncType}}	The type of the contact.
{{.ContactNumber}}	The number of the contact.
{{.ContactId}}	The unique ID of a contact.
{{.TicketId}}	The unique ID of the ticket created based on a call. **Note:** To get the ticket ID, you need to configure additional settings in the template. For more information, see Extended Configuration - Retrieve Ticket ID.
{{.CallDuration}}	The time between the call started and the call ended.
{{.Talk_Duration}}	The time between the call answered and the call ended (in a format of `hh:mm:ss`).
{{.Talk_Duration_Sec}}	The time between the call answered and the call ended (in seconds).
{{.Communication_Type}}	The direction of the call. **Possible value**: • `Inbound`: Inbound call. • `Outbound`: Outbound call.
{{.Call_Status}}	The processing status of the call. **Possible value**: • `Missed`: A call that was not answered. • `Completed`: A call that was successfully connected and completed.
{{.Call_Log_Status}}	The status of the call. **Possible value**: • `Missed Call`: A call that was not answered, • `Outgoing Call`: An outbound call that was initiated by the system. • `Incoming Call`: An inbound call that was received by the system. • `Voicemail`: A call that was not answered and was directed to voicemail.

Variable	Description
{{.CallDisposition}}	The disposition of the call. **Possible value**: • `MISSED`: The call was not answered. • `VOICEMAIL`: The call was sent to voicemail. • `BUSY`: The recipient was on another call and could not answer. • `ANSWERED`: The call was successfully answered. • `NO ANSWER`: The recipient did not answer the call.

Call journal scenario

This scenarios is used to log the call activities of associated users to the CRM / Helpdesk. The structure of the scenario and its key elements are outlined as below:

```
<Scenario Id="CallJournal" Type="REST">
   <Presets></Presets>
   <Parameters>
    <Parameter Name="EnableSubject" Value="1"></Parameter>
    <Parameter Name="Subject" Value="Extension Call"></Parameter>
    <Parameter Name="EnableDescription" Value="1"></Parameter>
    <Parameter Name="Description" Value="Call: {{.Time}} {{.Call_Log_Status}} from {{.Call_From}} to
{{.Call_To}} {{.Talk_Duration}}"></Parameter>
    <Parameter Name="EnablePlayCallRecording" Value="1"></Parameter>
    <Parameter Name="PlayCallRecording" Value="0"></Parameter>
    <Parameter Name="EnableForwardCallLogForAnsweredAgent" Value="0"></Parameter>
    <Parameter Name="ForwardCallLogForAnsweredAgent" Value="0"></Parameter>
   </Parameters>
   <Requests>
     <Request Name="CallJournal" Method="POST" Weight="0" ResponseType="application/json" RequestEncoding=""
URLFormat="https://www.api.example.com/v2/Calls" RefreshTime="" IsDelete="0">
     <Parameters>
        <Parameter Name="Data" Type="Body" Value="{&#xA;  &#34;data&#34;: [&#xA;    {&#xA;      &#34;
{{.Owner}}{{.WhoModule}}&#34;: {&#xA;        &#34;Description&#34;: &#34;{{.Description}}&#34;,&#xA;
&#34;Voice_Recording__s&#34;: &#34;{{.RecordPath}}&#34;,&#xA;        &#34;Call_Start_Time&#34;: &#34;
{{TimeFormat .StartTimeUnix &#39;yyyy-MM-ddTHH:mm:ss-z&#39; &#39;1&#39;}}&#34;,&#xA;
&#34;Subject&#34;: &#34;{{.Subject}}&#34;,&#xA;        &#34;Call_Type&#34;: &#34;Inbound&#34;,&#xA;
&#34;Call_Result&#34;: &#34;{{.Call_Log_Status}}&#34;,&#xA;        &#34;Call_Duration&#34;: &#34;
{{.Talk_Duration_Sec}}&#34;&#xA;      }&#xA;    }&#xA;  ]&#xA;}"></Parameter>
     </Parameters>
     <Outputs></Outputs>
     </Request>
     <Request Name="CallJournalUpdate" Method="PUT" Weight="0" ResponseType="application/json"
RequestEncoding="" URLFormat="https://www.api.example.com/v2/Calls/{{.CallId}}" RefreshTime="" IsDelete="0">
     <Parameters>
         <Parameter Name="Data" Type="Body" Value="{&#xA;  &#34;data&#34;: [&#xA;    {&#xA;      &#34;
{{.Owner}}{{.WhoModule}}&#34;: {&#xA;        &#34;Description&#34;: &#34;{{.Description}}&#34;,&#xA;
&#34;Voice_Recording__s&#34;: &#34;{{.RecordPath}}&#34;,&#xA;        &#34;Call_Start_Time&#34;: &#34;
```

```
{{TimeFormat .StartTimeUnix &#39;yyyy-MM-ddTHH:mm:ss-z&#39; &#39;1&#39;}}&#34;,&#xA;
&#34;Subject&#34;: &#34;{{.Subject}}&#34;,&#xA;        &#34;Call_Type&#34;: &#34;Inbound&#34;,&#xA;
&#34;Call_Result&#34;: &#34;{{.Call_Log_Status}}&#34;,&#xA;        &#34;Call_Duration&#34;: &#34;
{{.Talk_Duration_Sec}}&#34;,&#xA;        &#34;AI_Transcription&#34;: &#34;{{.AI_Transcription}}&#34;,&#xA;
&#34;AI_Summary&#34;: &#34;{{.AI_Summary}}&#34;,&#xA;        &#34;Call_Note&#34;: &#34;
{{.Call_Note}}&#34;&#xA;      }&#xA;    }&#xA;  ]&#xA;}"></Parameter>
     </Parameters>
     <Outputs></Outputs>
     </Request>
   </Requests>
</Scenario>
```

Here is a detailed explanation of the specific elements within the scenario.

Scenario element

The `<Scenario>` element has the following attributes:

Attribute	Description
Id	The unique identifier of the scenario, the value should be fixed as `CallJournal`.
Type	The type of the scenario, the value is `REST`, which means the scenario will execute a REST API using an HTTP request.

Parameter element

Attribute	Description
Name	The name of the configuration settings used in the scenario.
Value	The value of the configuration settings.

Configuration settings

Settings	Description
EnableSubject	Whether the call log subject is configurable in the integration. **Valid value**: • `0`: Disable. • `1`: Enable.
Subject	Specify the default value of the subject. **Note:** The default value is `Extension Call`, you can also specify the subject using the supported variables.

Settings	Description
EnableDescription	Whether to the call log description is configurable in the integration. **Valid value**: • `0`: Disable. • `1`: Enable.
Description	Specify the default value of the description. **Note:** The default value is `{{.Time}} {{.Communication_Type}} {{.Call_Status}} - from {{.Call_From}} to {{.Call_To}} {{.Talk_Duration}}`. For more supported variables, see the parameter variable list.
EnablePlayCallRecording	Whether the feature to play call recordings within the CRM/Helpdesk is configurable in the integration. **Valid value**: • `0`: Disable. • `1`: Enable.
PlayCallRecording	Specify the default status of the call recording playback option. **Valid value**: • `0`: Disable. • `1`: Enable.
EnableForwardCallLogForAnsweredAgent	Whether the option to prevent logging missed calls for queue or ring group calls to non-answering agents' CRM/Helpdesk is configurable in the integration.
ForwardCallLogForAnsweredAgent	Specify the default status of the option to prevent logging missed calls for queue or ring group calls to non-answering agents' CRM/Helpdesk. **Valid value**: • `0`: Disable. • `1`: Enable. Call logs for queue and ring group calls are synchronized only to the CRM/Helpdesk of the agent who answered the call, while missed call logs for the same call are not synchronized to the CRM of agents who did not answer.

Request element

The `<Request>` element has the following attributes:

Attribute	Description
Name	The name of the request, which helps to identify the request within the scenario. **Valid value**: • `CallJournal`: The call journaling creation request to create a new call log in the CRM/Helpdesk. • `CallJournalUpdate`: The call journaling update request to update an existing call log with additional information.
Method	The HTTP method to be used for the request. **Valid value**: • `POST` • `PUT`
ResponseType	The expected response format for the request, which should be in **JSON** format (`application/json`).
RequestEncoding	How the request should be encoded. In this scenario, the value is empty.
URLFormat	The request URL of the API endpoint.

Parameter element

The `<Parameter>` element nested within the `<Request>` element defines the parameters included in the request body.

Attribute	Description
Name	The name of the parameter, which should be fixed as `Data`.
Type	The type of the parameter, which should be fixed as `Body`.
Value	Defines the required parameters to be passed through the request body. The parameter values can be retrieved from the pre-defined variables in the PBX system. **Note:** The format should follows XML parameter standards. For example, if a line break is required, you must use the escape character `
` to represent a newline.

Variables

Note: Only required variables are listed here. If you need additional parameters and variables, see Extended Configurations - Retrieve Additional Variables.

XML Descriptions for Integration Template

Yeastar P-Series Cloud Edition - CRM / Helpdesk Integration Template

Variable	Description
Variables obtained from the `AuthMethod` scenario	The variables captured from the authentication credentials input during the authentication process of integration.
Target variables obtained from the follow-up requests	The target variables extracted from the responses of the follow-up requests.
{{.Subject}}	The subject of the call. This variable value is retrieved from the `Subject` parameter.
{{.Description}}	The description of the call. The variable value is retrieved from the `Description` parameter.
{{.LinkedId}}	The unique ID of the call.
{{.Time}}	The time the call was made or received (in PBX system's time format).
{{.StartTimeUnix}}	The Unix timestamp of the call start time.
{{.EndTimeUnix}}	The Unix timestamp of the call end time.
{{.Call_From}}	The name and number of the calling party that initiated the call. **Format**: `Caller's name <Caller's number>`.
{{.CallerName}}	Caller's name.
{{.CallerNumber}}	Caller's number.
{{.CalleeName}}	Callee's name.
{{.CalleeNumber}}	Callee's number.
{{.CrmId}}	The unique ID of the CRM/Helpdesk user associated with a PBX extension.
{{.ExtensionFirstName}}	The first name of the extension user.
{{.ExtensionLastName}}	The last name of the extension user.
{{.ExtensionEmail}}	The email of the extension user.

XML Descriptions for Integration Template

Yeastar P-Series Cloud Edition - CRM / Helpdesk Integration Template

Variable	Description
{{.ContactSyncType}}	The type of the contact.
{{.ContactNumber}}	The number of the contact.
{{.ContactId}}	The unique ID of a contact.
{{.TicketId}}	The unique ID of the ticket created based on a call. **Note:** To get the ticket ID, you need to configure additional settings in the template. For more information, see Extended Configuration - Retrieve Ticket ID.
{{.CallDuration}}	The time between the call started and the call ended.
{{.Talk_Duration}}	The time between the call answered and the call ended (in a format of `hh:mm:ss`).
{{.Talk_Duration_Sec}}	The time between the call answered and the call ended (in seconds).
{{.Communication_Type}}	The direction of the call. **Possible value**: • `Inbound`: Inbound call. • `Outbound`: Outbound call.
{{.Call_Status}}	The processing status of the call. **Possible value**: • `Missed`: A call that was not answered. • `Completed`: A call that was successfully connected and completed.
{{.Call_Log_Status}}	The status of the call. **Possible value**: • `Missed Call`: A call that was not answered, • `Outgoing Call`: An outbound call that was initiated by the system. • `Incoming Call`: An inbound call that was received by the system. • `Voicemail`: A call that was not answered and was directed to voicemail.

Variable	Description
{{.CallDisposition}}	The disposition of the call. **Possible value**: • `MISSED`: The call was not answered. • `VOICEMAIL`: The call was sent to voicemail. • `BUSY`: The recipient was on another call and could not answer. • `ANSWERED`: The call was successfully answered. • `NO ANSWER`: The recipient did not answer the call.
{{.EnableCallRecording}}	Whether the call recording is enabled. **Possible value**: • `0`: Disabled. • `1`: Enabled.
{{.AI_Transcription}}	AI-generated call transcript content.
{{.AI_Summary}}	AI-generated call summary content.
{{.Call_Note}}	Call note content.

Chat journal scenario

This scenario is used to synchronize the chat messages between associated users and the contacts to the CRM / Helpdesk. The structure of the scenario and its key elements are outlined as below:

```
<Scenario Id="ChatJournal" Type="REST">
  <Presets></Presets>
  <Parameters>
     <Parameter Name="EnableSubject" Value="1"/>
     <Parameter Name="Subject" Value="PhoneSystem External Chat Session"/>
     <Parameter Name="EnableCreateType" Value="0"/>
     <Parameter Name="CreateType" Value=""/>
  </Parameters>
  <Requests>
    <Request Name="ChatJournalDefault" Method="POST" Weight="0" ResponseType="application/json"
RequestEncoding="" URLFormat="https://www.api.example.com/crm/v7/Tasks" RefreshTime="" IsDelete="0">
      <Parameters>
        <Parameter Name="Data" Type="Body" Value="{ "data": [ { "Owner": { "id": "{{.OwnerId}}" }, "Who_Id":
"{{.ContactEntityId}}", "Subject": "{{.Subject}}", "Due_Date": "{{ TimeFormat .StartTimeUnix "yyyy-MM-dd"
"0" }}", "Closed_Time": "{{ TimeFormat .EndTimeUnix "yyyy-MM-ddTHH:mm:ss.000Zz" "0" }}", "Status":
"Completed", "Priority": "Normal", "Description": "Message Channel Type:
{{.Channel_Display}}\n\n{{.Messages}}" } ] }"/>
```

```
      </Parameters>
      <Outputs/>
    </Request>
  </Requests>
</Scenario>
```

Here is a detailed explanation of the specific elements within the scenario.

Scenario element

The `<Scenario>` element has the following attributes:

Attribute	Description
Id	The unique identifier of the scenario, the value should be fixed as `ChatJournal`.
Type	The type of the scenario, the value is `REST`, which means the scenario will execute a REST API using an HTTP request.

Parameter element

Attribute	Description
Name	The name of the configuration settings used in the scenario.
Value	The value of the configuration settings.

Configuration settings

Attribute	Description
EnableSubject	Whether the subject of the synced record is configurable in the integration. **Valid value**: • `0`: Disable. • `1`: Enable.
Subject	Specify the default value of the subject. **Note:** The default value is `PhoneSystem External Chat Session`, you can also specify the subject using the supported variables.
EnableCreateType	Reserved field (not applicable).
CreateType	Reserved field (not applicable).

Request element

The `<Request>` element has the following attributes:

Attribute	Description
Name	The name of the request, which helps to identify the request within the scenario. The value should be fixed as `ChatJournalDefault`.
Method	The HTTP method to be used for the request. **Valid value**: • `POST` • `PUT`
ResponseType	The expected response format for the request, which should be in **JSON** format (`application/json`).
RequestEncoding	How the request should be encoded. In this scenario, the value is empty.
URLFormat	The request URL of the API endpoint.

Parameter element

The `<Parameter>` element nested within the `<Request>` element defines the parameters included in the request body.

Attribute	Description
Name	The name of the parameter, which should be fixed as `Data`.
Type	The type of the parameter, which should be fixed as `Body`.
Value	Defines the required parameters to be passed through the request body. The parameter values can be retrieved from the pre-defined variables in the PBX system. **Note:** The format should follows XML parameter standards. For example, if a line break is required, you must use the escape character `
` to represent a newline.

Variavles

Note: Only required variables are listed here. If you need additional parameters and variables, see Extended Configurations - Retrieve Additional Variables.

XML Descriptions for Integration Template

Yeastar P-Series Cloud Edition - CRM / Helpdesk Integration Template

Variable	Description
Variables obtained from the `AuthMethod` scenario	The variables captured from the authentication credentials input during the authentication process of integration.
Target variables obtained from the follow-up requests	The target variables extracted from the responses of the follow-up requests.
{{.ContactNumber}}	The phone number of the CRM / Helpdesk contact.
{{.ContactEmail}}	The email address of the CRM / Helpdesk contact.
{{.ContactName}}	The name of the CRM / Helpdesk contact.
{{.ContactEntityId}}	The entity ID of the matched contact, lead, or account in the CRM / Helpdesk.
{{.ContactEntityType}}	The type of the matched entity (contact, lead, or account).
{{.QueueName}}	The name of the message queue. This is included when the chat is routed to a queue.
{{.Messages}}	The messages sent and received during the session, including time, senders, and message content.
{{.ExtensionNumber}}	The extension number of the extension or agent who handled the chat messages or closed the session. **Note:** If the session is closed automatically by the system, this value is set to `PBX System`.
{{.ExtensionFirstName}}	The first name of the extension or agent who handled the chat messages or closed the session. **Note:** If the session is closed automatically by the system, this value is set to `PBX System`.
{{.ExtensionLastName}}	The last name of the extension or agent who handled the chat messages **or** closed the session. **Note:** If the session is closed automatically by the system, this value is set to `PBX System`.
{{.ExtensionEmail}}	The email address of the extension or agent who handled the chat messages.
{{.ChatDuration}}	The duration of the chat session, recorded in `hh:mm:ss` format. To convert to total minutes (using the `ToMinutes` function): `{{ToMinutes .ChatDuration}}`

Variable	Description

{{.StartTimeUnix }}	The Unix timestamp (seconds) of the session start time.
	• To format as local time string (using the `TimeFormat` function):
	`{{ TimeFormat .StartTimeUnix "yyyy-MM-ddTHH:mm:ss.000Zz" "0" }}`
	• To format as UTC time string (using the `TimeFormat` function):
	`{{ TimeFormat .StartTimeUnix "yyyy-MM-ddTHH:mm:ss.000Zz" "1" }}`
	• To convert to milliseconds (using the `ToMillis` function):
	`{{ToMillis .StartTimeUnix}}`

{{.EndTimeUnix}}	The Unix timestamp (seconds) of the session end time.
	• To format as local time string (using the `TimeFormat` function):
	`{{ TimeFormat .EndTimeUnix "yyyy-MM-ddTHH:mm:ss.000Zz" "0" }}`
	• To format as UTC time string (using the `TimeFormat` function):
	`{{ TimeFormat .EndTimeUnix "yyyy-MM-ddTHH:mm:ss.000Zz" "1" }}`
	• To convert to milliseconds (using the `ToMillis` function):
	`{{TsSecToMilliSec .EndTimeUnix}}`

Extended configurations

In addition to the scenarios specified in the configuration page, you can also manually add custom configurations for supplementary use.

Note: Custom configurations in the template are preserved and will not be affected by changes made through the PBX web portal.

Set custom request headers

If custom request headers are needed, you can add parameter(s) with the type set to `Header`. This will add the headers to the request body.

As shown in the following example, a `X-Auth-Token` header is used to pass the API key. The `Father` attribute (fixed as `PopulateTemplateString`) groups a set of related parameters, while the `Type` (fixed as `Header`) indicates that the parameter is intended to be passed as an HTTP header. The `Name` and `value` specify the header's name and value. In this case, the header included in the request would be `X-Auth-Token:{{.api_key}}`.

```
 <Requests>
        <Request Name="contacts" Method="GET" ResponseType="application/json"
URLFormat="https://api.example.come/contacts/search?number={{.Phone}}">
          <Parameters>
            <Parameter Father="PopulateTemplateString" Name="X-Auth-Token" Type="Header" Value="
{{.api_key}}" />
          </Parameters>
          <Outputs>
           ...
          </Outputs>
        </Request>
```

Identify token errors

In the `AuthMethod` scenario, you can add a parameter `TokenInvalid` to help identify whether the issue in the request is related to the token or other underlying problems.

You can refer to the following example, and change the `Value` to the token-related error code or specific content. If the HTTP status code returned is not `200` and the response body contains the content specified in `Value`, the error will recognized as **TokenInvalid**.

```
   <Scenario Id="AuthMethod" Type="AUTH">
      <Presets />
      <Parameters>
        <Parameter Name="AuthMethod" Value="oauth2" />
        ...
        <Parameter Father="TokenErrorMap" Type="TokenInvalid" Name="TokenInvalid" Value="INVALID_TOKEN" />
        ...
      </Parameters>
      <Requests />
      <Responses />
    </Scenario>
```

Retrieve additional variables for global use

If you require additional variables beyond the provided ones, if you want to use certain common information globally during the integration process, you can define a `GetGlobalInfo` scenario and set up the related requests to retrieve the necessary values.

When the integration is completed or the PBX is started, requests set within the `GetGlobalInfo` scenario will be sent to the CRM/Helpdesk system, and the required variables will be stored in the `Output` elements, with their names directly corresponding to the variable names, making them accessible for reference in other scenarios (excluding the authentication scenario).
As shown in the following example, the `GetGlobalInfo` scenario contains requests for retrieving essential information `DepartmentId` and `PortalName`.

```
<Scenario Id="GetGlobalInfo" Type="REST">
      <Presets />
      <Parameters />
      <Requests>
        <Request Name="GetDepartments" Method="GET" ResponseType="application/json"
URLFormat="https://desk.example.{{.domain_suffix}}/api/v1/departments">
          <Parameters />
          <Outputs>
            <Output Name="DepartmentId" Path="data.#.id" />
          </Outputs>
        </Request>
        <Request Name="GetOrganizations" Method="GET" ResponseType="application/json"
URLFormat="https://desk.example.{{.domain_suffix}}/api/v1/organizations">
          <Parameters />
          <Outputs>
            <Output Name="PortalName" Path="data.#(portalName!="").portalName" />
          </Outputs>
        </Request>
      </Requests>
</Scenario>
```

Supplement contact query information

If the initial query for contact information does not return all the necessary information you need, further queries may be required using the returned contact ID or other contact information.

As shown in the example below, in a scenario of contact synchronizing, the initial query request does not return the desired phone numbers.

```
<Scenario Id="SyncContactAuto" Type="REST">
   <Presets />
   <Parameters>
     <Parameter Name="ContactUrlType" Value="specify_url_format" />
        ...
     </Parameters>
     <Requests>
       <Request Name="contacts" Method="GET" ResponseType="application/json"
URLFormat="https://{{.subdomain}}.example.com/api/v2/users?query=phone:{{UrlEncode .Phone}}&role[]=end-
user">
         <Parameters />
         <Outputs Next="GetIdentities">
            <Output Name="ContactsId" Path="users.#.id" />
            <Output Name="FirstName" Path="users.#.name" />
            <Output Name="BusinessNumber" Path="users.#.phone" />
            ...
            //The following phone number fields returns empty.
            <Output Name="BusinessNumber2" Path=""></Output>
            <Output Name="BusinessFaxNumber" Path=""></Output>
            <Output Name="MobileNumber" Path=""></Output>
            <Output Name="MobileNumber2" Path=""></Output>
            <Output Name="HomeNumber" Path=""></Output>
            <Output Name="HomeNumber2" Path=""></Output>
            ...
          </Outputs>
        </Request>
```

```
    </Requests>
  </Scenario>
```

In this case, you can perform a further query to retrieve the desired numbers by chaining scenarios, as described in the following steps:

1. **Add a Next Attribute in the original scenario**
   In the contact synchronizing scenario, add a `Next` attribute in the `<Outputs>` element of the initial query request. The attribute value must be the name of the desired request configured in the `Common` scenario. In this example, the request name is `GetIdentities`.

   ```
   <Outputs Next="GetIdentities">
   ```

2. **Create a Common scenario**
   Create a scenario named `Common` with a type of `REST` (both the name and type are fixed), then add a request named `GetIdentities` to query additional contact information using the variable `{{.ContactsId}}` obtained from the initial request.

   ```
   <Scenario Id="Common" Type="REST">
     <Presets></Presets>
     <Parameters></Parameters>
     <Requests>
       <Request Name="GetIdentities" Method="GET"
   URLFormat="https://{{.subdomain}}.example.com/api/v2/users/{{.ContactsId}}/identities"
   ResponseType="application/json">
         <Parameters></Parameters>
         <Outputs>
           <Output Name="BusinessNumber2" Path="identities.#(type=="phone_number")#|1.value"></Output>
           <Output Name="BusinessFaxNumber" Path="identities.#(type=="phone_number")#|2.value"></Output>
           <Output Name="MobileNumber" Path="identities.#(type=="phone_number")#|3.value"></Output>
           <Output Name="MobileNumber2" Path="identities.#(type=="phone_number")#|4.value"></Output>
           <Output Name="HomeNumber" Path="identities.#(type=="phone_number")#|5.value"></Output>
           <Output Name="HomeNumber2" Path="identities.#(type=="phone_number")#|6.value"></Output>
           <Output Name="HomeFaxNumber" Path="identities.#(type=="phone_number")#|7.value"></Output>
           <Output Name="OtherNumber" Path="identities.#(type=="phone_number")#|8.value"></Output>
         </Outputs>
       </Request>
     </Requests>
   </Scenario>
   ```

   Since the request is referenced via the `NEXT` attribute in the original scenario, the variable from the initial query will be passed to the request. The request will execute and backfill the `Outputs` into the initial request's response to complete the information.

Retrieve TicketID for further operations

In a Helpdesk integration template, you can configure additional settings to retrieve the ticket ID for subsequent operations, such as updating the ticket or referencing the variable in call journal.

For example, when the ticket is created before the call (i.e. when the phone rings), it might be missing some information. In this case, you can update the information for the created ticket by using the `TicketId` in the

corresponding call record later, as described in the following steps:

```
<Scenario Id="CreateNewTicket" Type="REST">
   <Presets />
   <Parameters>
      <Parameter Name="EnableSubject" Value="1" />
      ...
      <Parameter Name="NeedSyncContact" Value="1" />
    </Parameters>
    <Requests>
       <Request Name="CreateNewTicket" Method="POST" ResponseType="application/json"
URLFormat="https://{{.domain}}/api/v1/tickets">
         <Parameters>
          <Parameter Name="Data" Type="Body" Value="{"departmentId":"{{.DepartmentId}}","subject":"
{{.Subject}}","contactId":"{{.ContactId}}","phone":"{{.ContactNumber}}","description":"{{.Description}}"}"
/>
         </Parameters>
         <Outputs>
           <Output Name="TicketId" Path="id" />
         </Outputs>
        </Request>
        <Request Name="UpdateTicket" Method="POST" ResponseType="application/json"
URLFormat="https://{{.domain}}/api/v1/tickets/{{.TicketId}}/comments">
          <Parameters>
            <Parameter Name="Data" Type="Body" Value="{"isPublic":"true","content":"{{.Description}}"}" />
          </Parameters>
        <Outputs />
       </Request>
    </Requests>
</Scenario>
```

1. **Retrieve ticket ID**

   In the `CreateNewTicket` scenario, add an `<Output>` element to extract the ticket ID from the response of `CreateNewTicket` request.

   ```
   <Outputs>
     <Output Name="TicketId" Path="id" />
   </Outputs>
   ```

   This allows the system to store the `TicketId` for future use.

2. **Update the ticket**

   Add a request to update the ticket in the corresponding call record using the previously retrieved ticket ID.

   ```
   <Request Name="UpdateTicket" Method="POST" ResponseType="application/json"
   URLFormat="https://{{.domain}}/api/v1/tickets/{{.TicketId}}/comments">
      <Parameters>
        <Parameter Name="Data" Type="Body" Value="{"isPublic":"true","content":"{{.Description}}"}" />
      </Parameters>
      <Outputs />
   </Request>
   ```

   The ticket is updated with all necessary data after the initial creation.